

4.3 Web Application for Employment Opportunity Prediction

4.3.1 System Architecture and Implementation

The employment opportunity prediction system was implemented as a web application using the Next.js framework, leveraging TypeScript for enhanced code quality and maintainability. The system follows a client-server architecture where the client-side handles user interactions and result presentation, while the server-side manages data processing, model inference, and database interactions through Next.js API routes.

The technology stack incorporated React components from shadcn UI to ensure a consistent and accessible user interface. Supabase provided backend-as-a-service capabilities, including PostgreSQL database support and authentication services. Prisma ORM facilitated type-safe database interactions, while the Vercel AI SDK enabled the generation of AI-driven insights based on model predictions and user profiles.

The web application comprises 84 distinct UI components organized into 8 main page components with a consistent layout structure. These components work in concert to provide a seamless user experience from initial profile submission through to the presentation of job role predictions and personalized insights.

The deployment architecture utilized Vercel's edge network to ensure low-latency access globally, with the Random Forest model serialized using pickle and deployed as a serverless function. This approach allowed for efficient scaling while maintaining prediction response times under 500ms for typical user profiles. Database operations were optimized through connection pooling and strategic indexing, resulting in consistent sub-second query performance even under load testing with 100 concurrent users.

[Figure 4.7: System Architecture Diagram]

This diagram illustrates the complete system architecture with client-side components (UI modules, state management), server-side elements (API routes, model inference, database integration), and external services (job search APIs, authentication). Component relationships are shown with directional arrows indicating data flow between system elements.

4.3.2 Resume Processing Capabilities

The system demonstrated robust resume processing capabilities, handling multiple file formats including PDF, DOCX, and image files. For PDF documents, text extraction was performed using PyMuPDF (fitz), while python-docx was employed for Microsoft Word documents. Image-based resumes underwent optical character recognition (OCR) using Pytesseract, achieving a 94.2% accuracy rate for clear images.

The resume parsing module employed a hybrid approach combining rule-based extraction with machine learning classification. Named entity recognition (NER) was implemented using spaCy to identify key entities such as organization names, job titles, and dates. This was supplemented with custom pattern matching rules to capture education credentials and domain-specific terminology. The system maintained high extraction accuracy across different resume layouts and formats, with performance metrics shown in Table 4.7.

Table 4.7: Resume Information Extraction Performance by Category

Information Category	Extraction Accuracy	Precision	Recall
Educational Institutions	96.3%	97.1%	95.4%
Degree/Certification	94.7%	95.2%	94.1%
Job Titles	93.8%	95.8%	91.7%
Companies/Organizations	95.4%	96.3%	94.6%
Skills and Technologies	92.3%	93.7%	90.8%
Dates and Time Periods	97.6%	98.2%	97.0%
Overall	94.2%	95.4%	93.0%

After extraction, users could review and modify the automatically extracted information through an intuitive skill selection interface, ensuring accurate profile representation. The system also implemented intelligent deduplication to prevent redundant skill entries and normalized variations in terminology (e.g., "Machine Learning" and "ML" were recognized as equivalent).

[Figure 4.8: Skill Selection Interface]

This visualization shows the interactive skill selection component where users can review automatically extracted skills from their resume and manually add, modify, or remove skills to ensure accurate representation of their capabilities. The interface uses toggleable skill chips with different colors representing skill categories.

4.3.3 Job Recommendation and Insight Generation

The inference pipeline integrated the Random Forest prediction model with LLM-based insight generation to provide comprehensive career guidance. The integration was achieved through a custom middleware layer that handled preprocessing, model invocation, and post-processing of predictions. The deployed model utilized the optimized hyperparameters identified during the development phase ($n_estimators=200$, $max_depth=20$, $min_samples_split=4$, $min_samples_leaf=2$, $max_features='sqrt'$, $criterion='gini'$).

The complete pipeline consisted of three main stages:

- 1. Preprocessing:** Converting user input (either from resume parsing or manual entry) into the numerical format required by the Random Forest model, applying the same transformations used during model training. This included one-hot encoding of categorical variables, ordinal encoding of education levels, and binary encoding of skills.
- 2. RF Prediction:** Generating the top 4-7 most probable job roles based on the preprocessed user profile, along with confidence scores for each prediction. The number of recommendations was dynamically adjusted based on the confidence distribution, with more recommendations provided when multiple roles had similar probability scores.
- 3. AI-Driven Insights:** Leveraging the Vercel AI SDK to generate personalized insights including strengths assessment, skill development recommendations, and career advice based on the

predicted job roles and user profile. The prompt engineering for this component was carefully optimized to produce consistent, relevant insights while avoiding generic advice.

To mitigate potential biases in the model recommendations, a bias detection and correction mechanism was implemented that analyzed prediction patterns across different demographic groups during development and applied appropriate adjustments to ensure fairness in the production system.

[Figure 4.9: Inference Pipeline Architecture]

This diagram illustrates the three-stage inference pipeline, showing the data flow from initial user input through preprocessing, Random Forest prediction, and LLM-based insight generation, culminating in the final results presented to the user. The diagram uses directional arrows to show data transformation between stages.

The system successfully integrated web search functionality to retrieve current job postings related to the predicted roles, providing users with actionable employment opportunities. This feature achieved an 87.4% relevance rate for top search results and 92.1% geographic precision, with an average retrieval time of 3.2 seconds. The job search implementation utilized a combination of direct API integration with major job platforms and custom web scraping with appropriate rate limiting and caching strategies to ensure reliable performance.

4.3.4 System Evaluation and User Experience

User testing was conducted with a sample of 30 participants representing diverse professional backgrounds. Participants were recruited across different career stages (entry-level, mid-career, senior) and various industries to ensure comprehensive evaluation. The testing protocol included both directed tasks and open exploration, followed by a detailed feedback questionnaire and semi-structured interviews.

The system achieved an overall satisfaction rating of 4.4 out of 5, with particularly strong scores for UI design (4.6/5) and job posting relevance (4.4/5). Resume parsing accuracy and job role relevance also received positive ratings of 4.2/5 and 4.3/5 respectively. The highest-rated feature was the skill development recommendations (4.7/5), while the lowest-rated was handling of unconventional career paths (3.8/5).

Qualitative feedback highlighted several key strengths:

- "The system provided job recommendations I hadn't considered but that aligned well with my skillset."
- "The skill gap analysis was particularly helpful in identifying targeted areas for professional development."
- "Integration of actual job listings saved significant time compared to my usual job search process."

When benchmarked against two existing job recommendation platforms (anonymized as Platform A and Platform B for this study), the system demonstrated competitive advantages in skill extraction precision (92.3% vs. 86.7% and 89.4%), job relevance scores (87.6% vs. 82.3% and 84.1%), and processing time (4.3s vs. 7.8s and 5.2s). The system also offered superior resume format support and real-time job integration compared to the benchmark platforms.

Table 4.8: Comparative Performance Evaluation Against Existing Platforms

Metric	Our System	Platform A	Platform B
Resume format support	Multiple (PDF, DOCX, Image)	Limited (PDF, DOCX)	PDF only
Skill extraction precision	92.3%	86.7%	89.4%
Job relevance score	87.6%	82.3%	84.1%
Processing time (avg)	4.3s	7.8s	5.2s
Real-time job integration	Yes	Yes	No
Personalized skill recommendations	Yes	Limited	No
Explanation for recommendations	Detailed	Basic	None
Multi-language support	Limited	Yes	No
Accessibility compliance	WCAG 2.1 AA	Partial	Partial

[Figure 4.10: User Experience Ratings]

This bar chart displays the average user ratings for different aspects of the system on a scale of 1-5, including ease of use (4.5), resume parsing accuracy (4.2), job role relevance (4.3), job posting relevance (4.4), UI design (4.6), and overall satisfaction (4.4), with error bars showing standard deviations.

4.3.5 Hybrid Approach Effectiveness

The integration of Random Forest with Large Language Models proved highly effective, combining the statistical rigor of traditional machine learning with the contextual understanding of LLMs. This hybrid approach successfully addressed the limitations of each method when used in isolation, providing both accurate predictions and meaningful explanations.

The synergy between these approaches resulted in several key improvements:

- 1. Temporal Adaptation:** The LLM component's ability to interpret recent developments in professional terminology allowed the system to remain relevant despite potential training data obsolescence. This was particularly evident in rapidly evolving fields like AI and blockchain, where new role descriptions emerge regularly.
- 2. Explanation Quality:** LLM-generated explanations effectively connected user skills to job requirements in a natural language format that would be difficult to achieve with Random Forest alone. These explanations achieved a human-rated relevance score of 4.6/5 compared to 3.2/5 for template-based explanations in an ablation study.
- 3. Edge Case Handling:** For profiles that didn't clearly align with common patterns in the training data, the LLM component provided more nuanced analysis and prevented the system from defaulting to overly generic recommendations. This improved handling of non-traditional career paths and interdisciplinary backgrounds.

4. **Contextual Skill Interpretation:** While the Random Forest model treated skills as binary features, the LLM component could interpret skill proficiency levels and recency from contextual information in the resume, providing more nuanced recommendations.

The cost-benefit analysis of this hybrid approach showed that while it increased computational requirements by approximately 60% compared to a pure Random Forest implementation, it reduced the need for frequent model retraining and significantly improved user satisfaction metrics, resulting in a favorable overall value proposition.

[Figure 4.11: Performance Comparison Between Random Forest Only and Hybrid Approach]

This visualization compares key performance metrics between the standalone Random Forest model and the hybrid approach (RF + LLM), using grouped bar charts to show the improvement across accuracy, precision, recall, and F1-score, with the hybrid approach consistently showing higher values.

4.3.6 Implications and Future Directions

The development and evaluation of this employment opportunity prediction system has several important implications for career guidance and workforce development:

1. **Personalization Over Generalization:** The success of the hybrid approach demonstrates that effective job matching requires both broad pattern recognition and personalized contextual understanding. This challenges the one-size-fits-all approach common in many existing job recommendation systems.
2. **Dynamic Skill Valuation:** The system's ability to adapt to emerging skills suggests that employment opportunity prediction must be viewed as a dynamic process rather than a static classification problem. This aligns with the rapidly evolving nature of professional requirements in the contemporary job market.
3. **Transparency and Explainability:** The positive user response to recommendation explanations highlights the importance of transparent AI systems in career guidance, where decisions have significant personal and professional impact. This finding supports the broader movement toward explainable AI in high-stakes domains.
4. **Bridging Education-Employment Gaps:** By identifying both skill strengths and potential gaps for desired roles, the system provides valuable guidance for targeted skill development, potentially improving employment outcomes through focused upskilling. This has implications for educational institutions seeking to align curricula with labor market demands.
5. **Accessibility and Inclusion:** The system's ability to process multiple resume formats and provide insights regardless of document structure promotes greater accessibility to career guidance, potentially benefiting candidates from diverse backgrounds and preparation levels.

Several limitations of the current implementation present opportunities for future enhancement:

1. **Longitudinal Validation:** While the system shows promising results in immediate testing, longitudinal studies tracking actual employment outcomes of users would provide stronger validation of the system's efficacy.

2. **Expanded Language Support:** The current implementation has limited support for non-English resumes, presenting an opportunity for improvement through multilingual NLP capabilities.
3. **Career Progression Modeling:** Future iterations could incorporate temporal modeling to suggest not just immediate job matches but optimal career progression paths over time.
4. **Regional Labor Market Integration:** Enhancing the system with region-specific labor market data would improve the relevance of recommendations in different geographic contexts.
5. **Continuous Learning Framework:** Implementing a feedback loop that incorporates user outcomes to continuously refine the underlying models would enhance system performance over time.

The web application successfully demonstrates how machine learning models and large language models can be effectively integrated within a user-friendly interface to provide valuable employment guidance, connecting individuals with relevant opportunities based on their unique combination of education, skills, and experience. This work contributes to both the technical literature on hybrid AI systems and the practical application of machine learning to workforce development challenges.